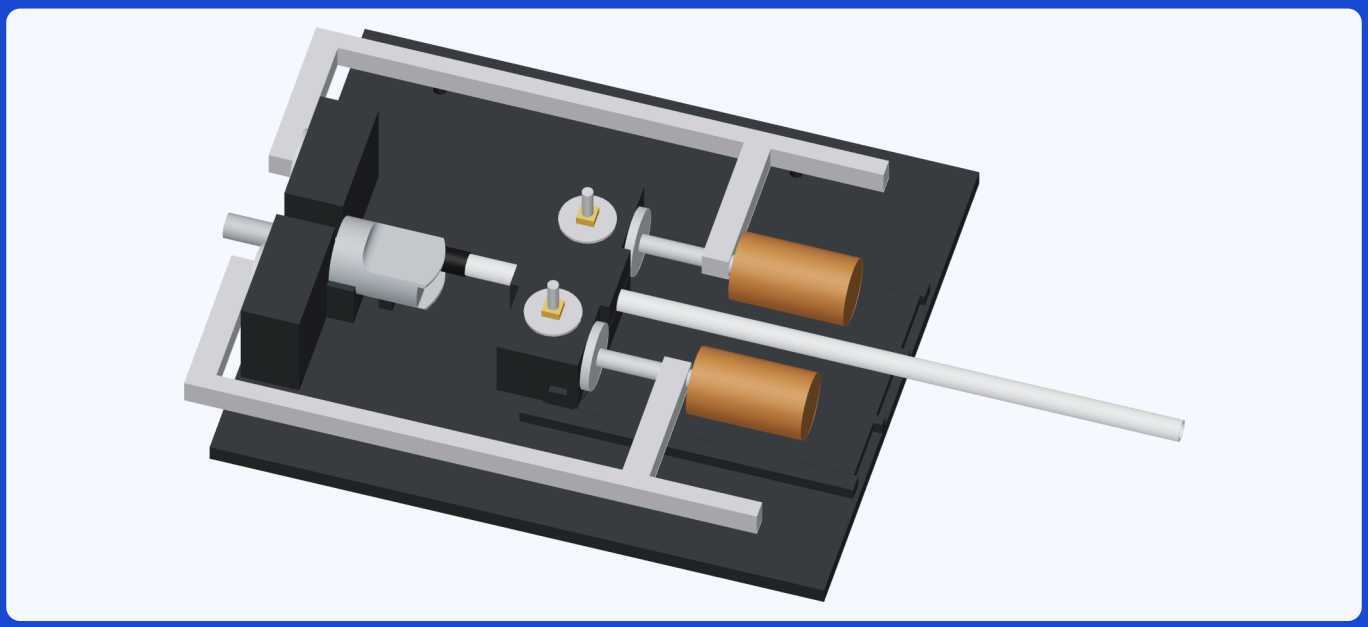
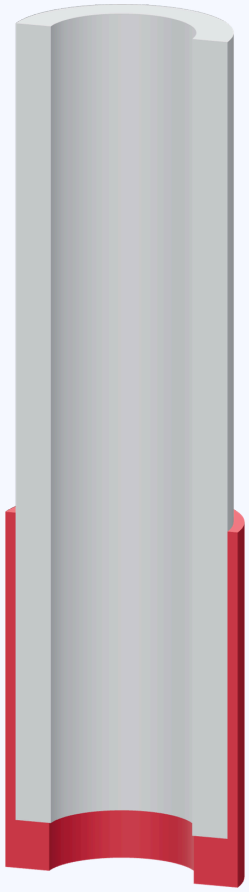
Tube insertion jig

Build the T handle and guided bench fixture. Fit the TPU thimble onto the soda tube, then use two metal screw clamps to insert the sleeved tube into the bare Touch-Flo.



The tube arrives sleeved

Two separate operations: stretch the thimble onto the tube outside the donor, then insert that sleeved end into the metal port.



The square-cut tube end rests on the thimble's annular cap. The cap hole remains open into the tube bore.

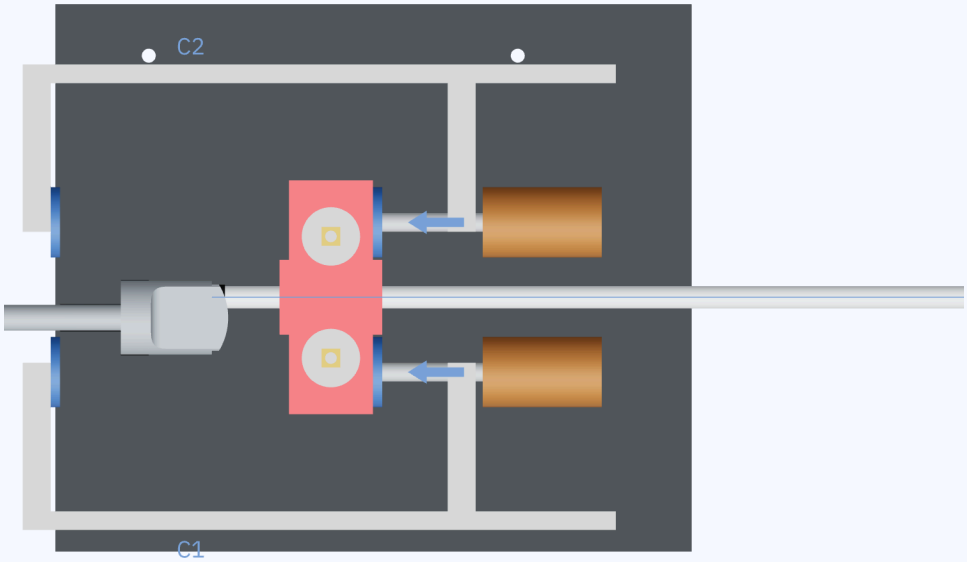
1. Attach the accepted printed faucet lever to the bare Touch-Flo. Leave it **down at rest, unpressed**.
2. Use the T handle to hold the tube while fitting the TPU thimble evenly around its end.
3. Place the bare Touch-Flo in the nest and press the **sleeved tube** into its empty top port.
4. Release the fixture. The faucet lever and sleeved tube remain attached when the donor enters the faucet shell.

THE THIMBLE GOES ONTO THE TUBE FIRST

TPU slides against the metal far more easily than against LLDPE. Putting the thimble into the donor first can bunch it under the entering tube.

Two metal force loops

Each metal clamp, C1 and C2, bears on an outboard contact beside the tube. Both contacts act on one guided grip; keep that grip square as it advances.



Top view: C1 and C2 connect their fixed pads at the donor block to their screw pads at the coral moving grip. Blue shows all four pad regions, the tube axis and screw-advance direction; arrow length does not represent force.

THROUGH THE FIXTURE	CONTACT
Each metal loop	The screw and moving pad load the grip; the fixed pad backs the donor reaction block. Each steel bar joins its fixed and sliding jaws to close the reaction path.
Grip to tube	Jaw pockets locate the TPU liners around the bare LLDPE tube. The tube and thimble transfer insertion resistance from the metal bore back to the grip.
Donor reaction	The metal base annulus rests against the donor reaction block. The arches, plunger and attached faucet lever remain clear.

Coral shows the part added or moved; existing prints are black, metal is grey, nuts are brass and alignment marks are blue. Pictures show nominal geometry. Do not measure a picture.

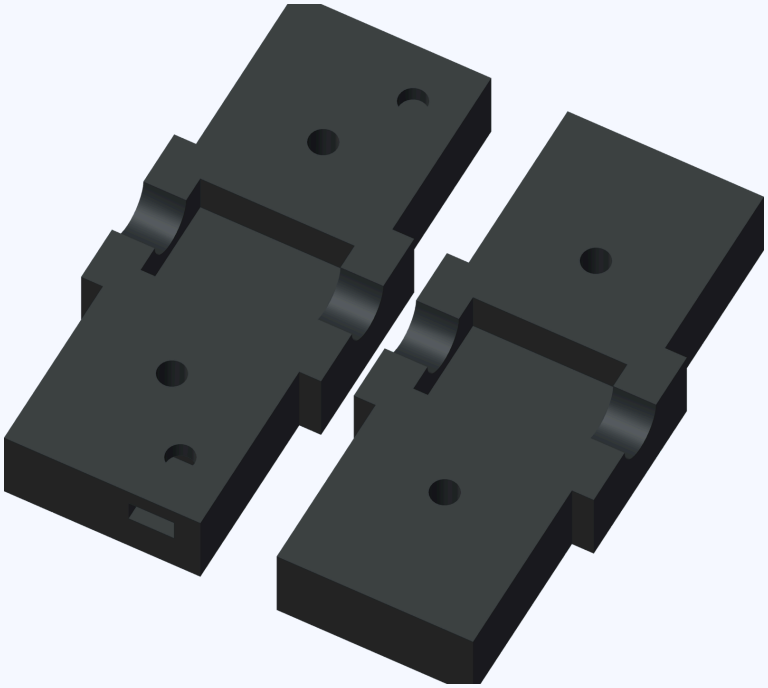
CHECK BOTH CONTACTS

Two pressure points describe where the screws act. They do not establish equal force sharing or a load rating.

PRINT · OPERATION

Print the centred grip

The replacement jaw plate uses Mark2's left 0.4 mm nozzle. Reuse the existing base, carrier and support puck. The manufacturing STLs carry their bed orientation.



D and E centred jaws, with flat backs on the bed and liner pockets upward.

PART	QTY	ORIENTATION
A Base and donor reaction block	1	Broad bottom down.
B Guided carrier	1	Flat support face up; feet down.
D / E Lower / upper jaws	1 each	Flat backs down; liner pockets up.
G Cap-support puck	1	Recess up.

`tube-insertion-rigid-base-h2c.3mf`
`tube-insertion-centered-grip-petgf-left04-z004-mark2.3mf`

Print D, E and two F liners for this grip. Keep the saved Tree(auto) / default support settings: D has four supports to remove from its underside key pockets and side-entry nut pockets. The matching `.gcode.3mf` archives are the prepared native slices.

Print the finished grip shape

Each liner is a thick block with a smooth half-round channel. Each liner stands 1 mm proud of its pocket, providing closing reserve around the tube.



Two copies of F, already shaped to fit the tube. Each flat backing seats directly in a rigid jaw pocket.

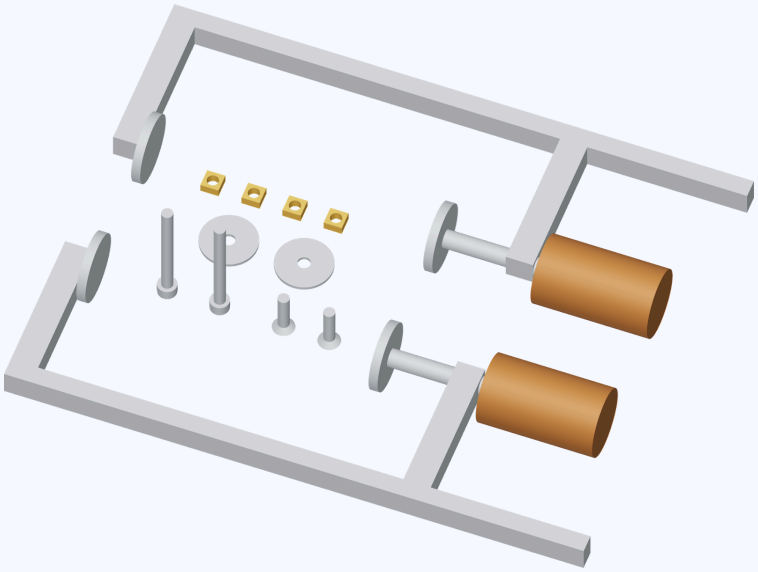
ITEM	PRINT
Grip liner	2 copies, TPU 85A. Use the prepared manufacturing STL orientation and saved project.
Print pose	Tube axis vertical; 28 mm tall; 9 × 20 mm footprint. Ø7.75 mm channel; 5.125 mm minimum backing.
Saved project	Mark1: right 0.6 mm nozzle, Z trim +0.18 mm. `tube-insertion-centered-liners-tpu85a-right06-z018-mark1.3mf`
Before assembly	Remove strings from the smooth contact channels and backing edges. Preserve the full backing and round gripping surface.

THE GRIP LINER AND WATER SEAL ARE DIFFERENT PARTS

These thick liners belong to the removable tool. The thin TPU 85A thimble belongs to the faucet's wet path and retains its existing geometry and print recipe.

Two clamps and four bolts

The complete metal screw clamps drive the common grip. Two stocked bolts close its jaws; two more join the lower jaw to the carrier.



Purchased clamps C1/C2, two M5 × 50 grip screws, two M5 × 20 carrier screws, four square nuts and two washers.

HARDWARE	QTY	WHERE
C1 / C2 BESSEY GSCC2.506 6-inch opening, 2.5-inch throat	2	Complete metal screw clamps at the two outboard contacts.
M5 × 50 socket-head screw	2	Grip-jaw closure.
M5 × 20 DIN 7991 countersunk screw	2	B-to-D joint, from the underside.
M5 × 0.8 square nut, 8 mm across flats × 4 mm thick	4	2 grip nuts, 2 carrier-joint nuts.
M5 fender washer, Ø25 × 1.5 mm, Ø5.5 mm bore	2	On upper jaw E.
Owned 4-inch steel C-clamp	2	Base-to-bench hold-down.

BESSEY GSCC2.506 - verified Amazon Prime listing

$10.99 each; two clamps total $21.98. Checked 2026-10-10.

The M5 screws, nuts and washers are already stocked. Clamp pictures show a nominal catalogue form; verify each factory pad seats fully on the fixture before applying load.

What you need at hand

The fixture bolts together. Its stop setup uses the actual donor's port and two marks on the tube.

TOOL OR WORKPIECE	FOR
3 mm and 4 mm hex keys	3 mm for the carrier's DIN 7991 screws; 4 mm for the grip's socket-head screws and the depth-reading fallback.
Small wrench or pliers	Hold the 8 mm square nuts while closing the two grip jaws.
Digital caliper with depth blade	Measure the actual donor's annular port floor and locate the two tube marks.
Tube cutter	A square, clean cut on the 3/8-inch OD LLDPE soda tube.
Fine marker	A port-lip seat witness and a clamp-front position mark, measured from the bare tube's square-cut end.
Firm bench	Support the complete fixture with clearance for both metal bars and both wooden handles.
Two owned 4-inch steel C-clamps	Hold the base to the bench for two-handed screw operation. Existing mounting bolts can also use its clear holes.
Plastic scraper, small cutters	Remove strings and print remnants from rail faces, liner pockets and gripping channels.
Bare Touch-Flo with accepted faucet lever	The workpiece. Its top port is empty when the sleeved tube approaches.
3/8-inch OD LLDPE soda tube	The faucet's finish-matched tube, prepared to the faucet assembly guide's length.
Fresh TPU 85A thimble	The existing faucet water seal, fitted onto the tube outside the donor.

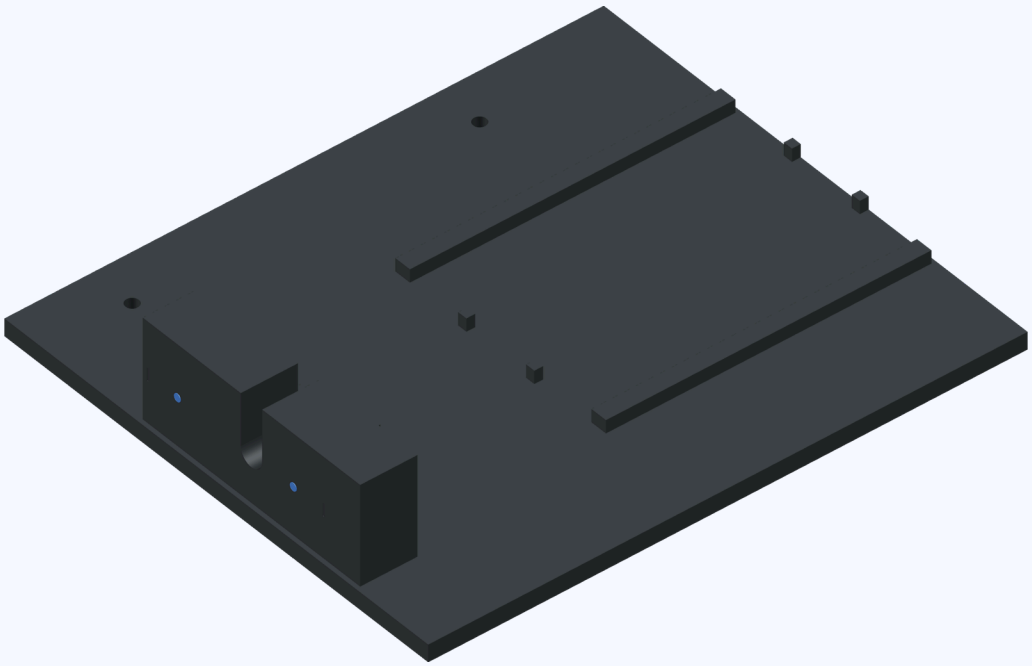
HOLD THE BASE STATIONARY

Secure the base before turning both screw handles. Keep the two complete screw clamps in their seating plane and leave each handle clear of the other clamp, the tube and the bench.

BUILD · OPERATION

Clean and mark base A

Base A combines the donor nest, reaction block, carrier guides and travel stops. Mark the two fixed-pad centres on the block's rear face.



Rear view of the reaction block: blue dots identify the two fixed-pad centres. The shank passes through the open notch; the donor annulus bears on the opposite face.

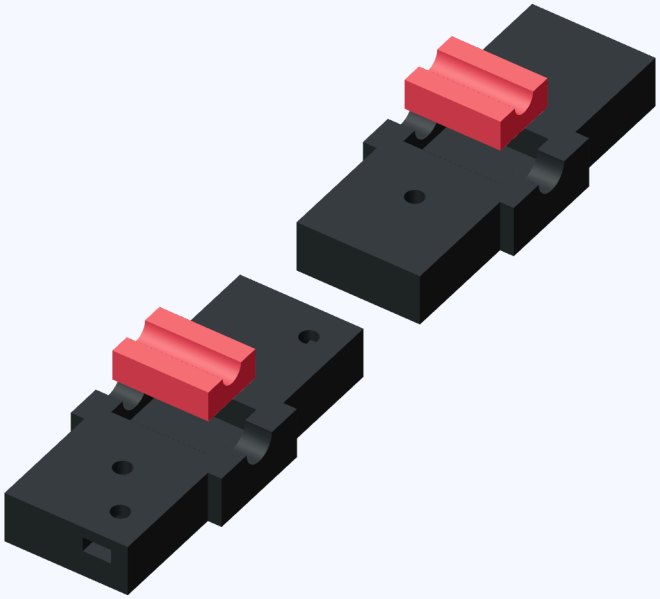
1. Remove print remnants and any supports generated by the prepared project. Clear the donor notch, guides and stop faces.
2. Remove strings and raised edges from the two fixed-pad regions and the donor annulus contact face. Preserve their flat bearing surfaces.
3. Clear the carrier path and guide faces. Keep the attached faucet lever and valve plunger clear of the fixture.
4. Use fine ink to mark two dots on the block's rear face, both **30 mm above the base underside**: the near dot is **36.75 mm from the front (-Y) edge**; the far dot is **19.25 mm from the back (+Y) edge** of the 120 mm-wide block.

FINISHED WHEN

The bearing faces and carrier path are clean. Both fixed-pad centres are marked on the flat rear face.

Seat one F in each jaw

The same thick liner fits each jaw. Its flat backing goes into the pocket and its smooth channel faces the tube.



D and E with their pockets facing up on the bench; one F liner poised over each pocket.

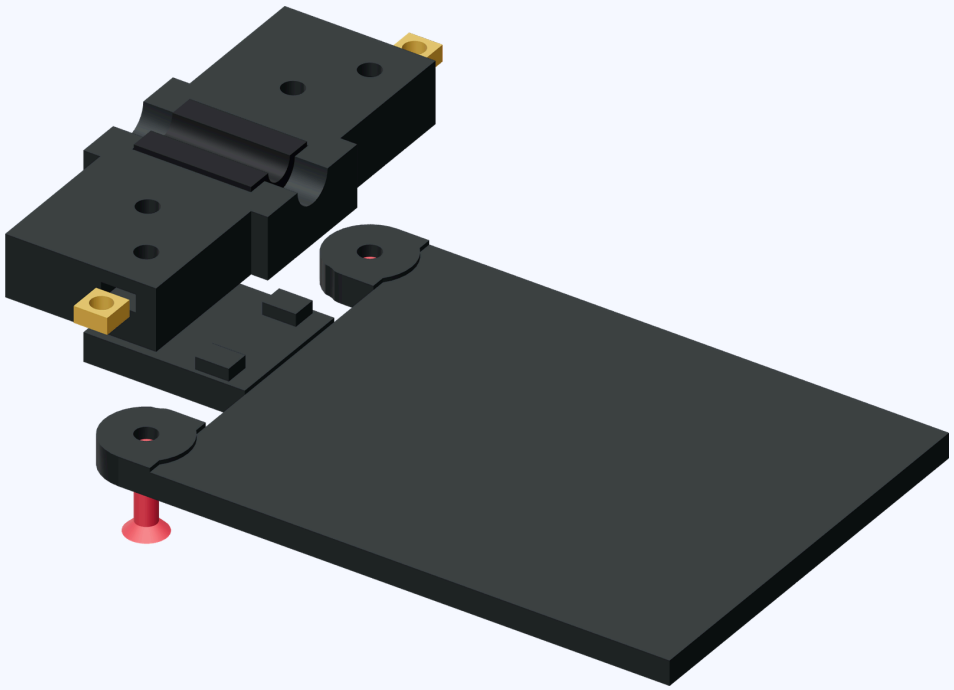
1. Remove D's four generated support bodies: two in its underside locating-key pockets and two in its side-entry square-nut pockets. Clear those openings before the B-to-D joint or nuts are fitted.
2. Lay lower jaw D and upper jaw E pocket-up on the bench.
3. Press one F liner fully into each pocket, flat backing against the pocket floor. It stands 1 mm proud of the pocket, with its half-round channel exposed.
4. Check each backing and end edge. Both backings must lie flat, without a trapped string or raised corner. Check that each liner's axial centre aligns with the two closure-screw holes.

FINISHED WHEN

The four support pockets are clear and both liners are seated flat. Their channels will face one another when the jaws close.

Bolt B to lower jaw D

Two countersunk screws join the carrier to the lower jaw. Side-loaded square nuts secure the joint before the carrier enters its guides.



Carrier B below lower jaw D, with two M5 × 20 DIN 7991 countersunk screws entering from B's underside and two M5 square nuts entering D's side pockets.

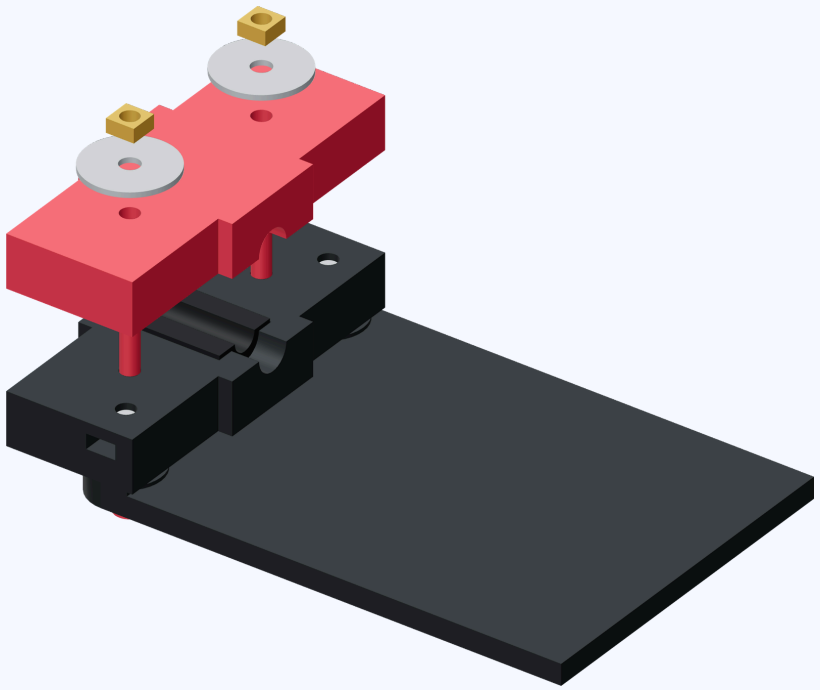
1. Load one M5 square nut into each lower-jaw side pocket, with its threaded hole aligned to the carrier screw axis.
2. Align D's underside pockets with the two raised locating keys on B. Keep D seated fully on the carrier support face.
3. Pass the two M5 × 20 countersunk screws upward through B and thread them into the square nuts.
4. Close the joint evenly. Confirm both countersunk heads seat at or below B's underside and the jaw does not move sideways on the carrier.

FINISHED WHEN

B and D form one detachable assembly. Both locating features are seated, both nuts are retained and the carrier underside is clear to slide.

Bolt the two jaws together

The wide rear wings provide a hand grip and two rear contact faces for the metal screw pads. The narrow front faces the sleeved tube end.



Lower jaw D already bolted to B, upper jaw E, two M5 × 50 screws, two Ø25 mm washers and two M5 square nuts.

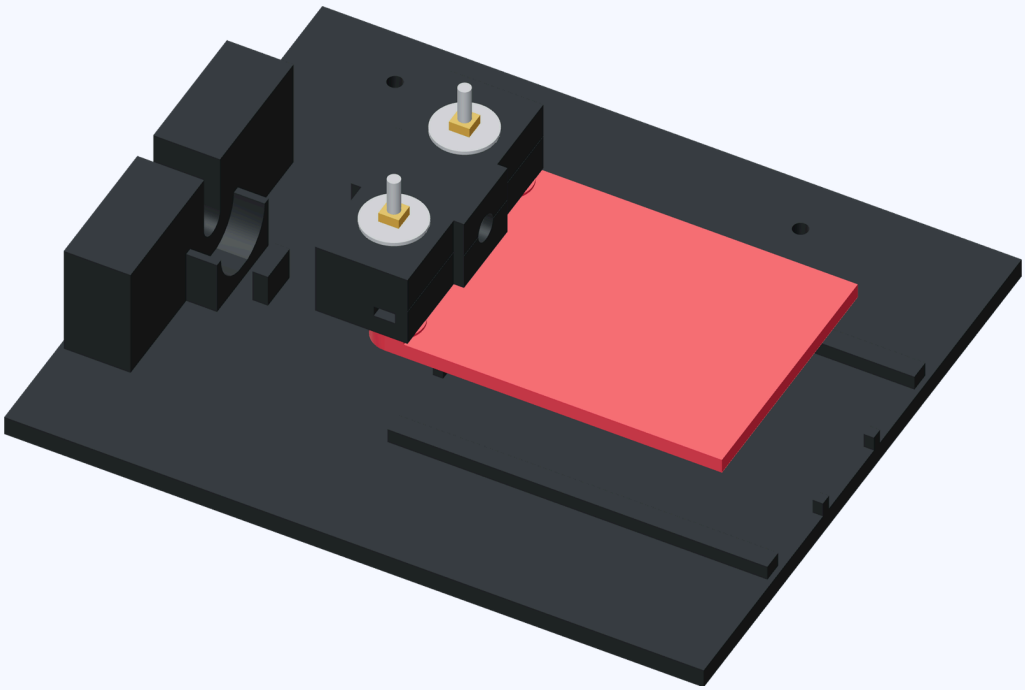
1. Pass one M5 × 50 screw upward through each of lower jaw D's two rear-wing holes.
2. Place upper jaw E over them with its liner channel facing the lower channel.
3. Put a Ø25 mm washer over each screw on the upper jaw, then thread on an M5 square nut.
4. Leave the two screws loose enough to open the jaws around a tube. When used, close the hard mating faces evenly; those faces limit the liner squeeze.
5. With the jaws closed, put two fine ink dots on their **rear face at the jaw seam**, each **18 mm inward from its outer wing edge**. These are the screw-pad centres, 32 mm each side of the tube axis.

FINISHED WHEN

The jaw halves open and close squarely, the washers sit fully on the rear wings, and both screw-pad centres are marked.

Slide the assembly into A

The T handle and carrier stay bolted together. The complete assembly lifts out for sleeve fitting and drops back into the base guides for insertion.



The joined carrier and grip above base A, with the carrier feet down and the sleeved-end side of the grip facing the donor nest.

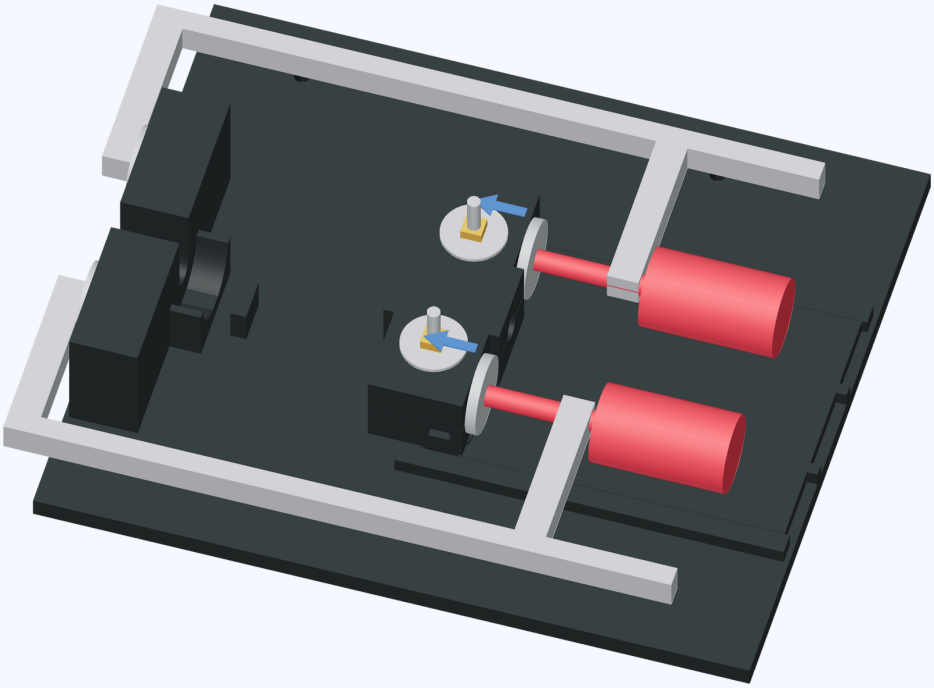
1. Set A flat and lower the joined B/D/E assembly between its guides, with the grip front toward the donor nest.
2. Move it unloaded to the paired front stops, then back through the 20 mm guided stroke.
3. Keep the carrier flat and square. Clear a string or raised print edge if it binds or lifts.
4. With the bare donor and attached faucet lever down at unpressed rest, check the complete unloaded stroke. The carrier, grip and clamps must leave the lever, arches and plunger clear.

FINISHED WHEN

The unloaded assembly moves freely, stays flat and reaches the paired stops together. Its full stroke leaves the attached faucet lever clear.

Preset travel without load

Use two complete BESSEY GSCC2.506 clamps. Set their available screw travel and handle clearance before placing the workpiece.



Complete nominal C1/C2 frames open outward. Pad opening is 142.5 mm at the 20 mm retracted position and 122.5 mm at the forward limit.

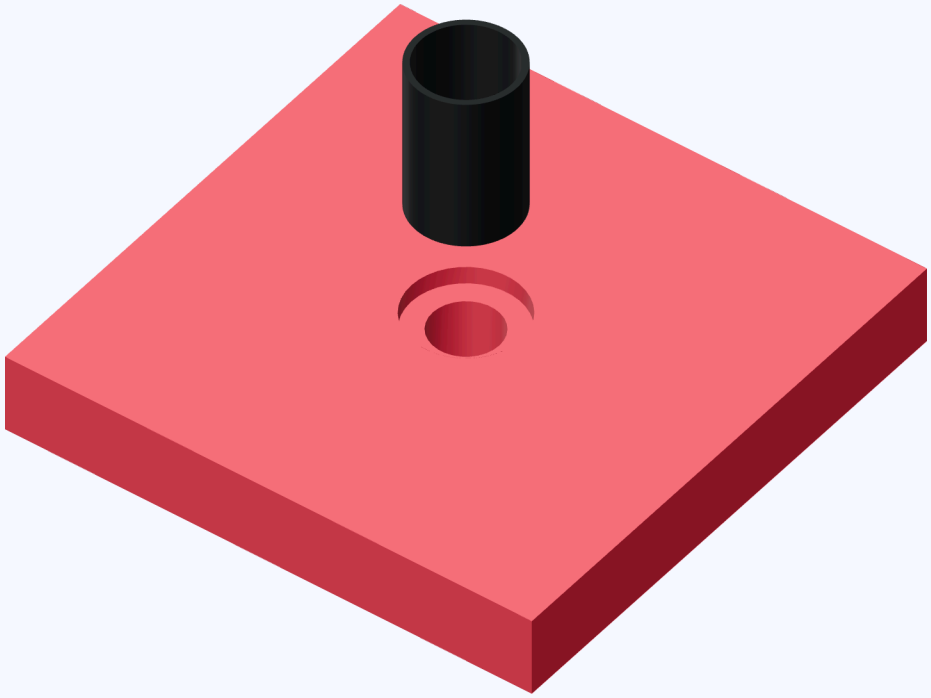
1. Inspect both clamps unloaded. Each sliding jaw must hold position, each screw must turn freely and each swivel pad must be able to seat flat.
2. Set each sliding jaw for a 142.5 mm starting pad opening. Advance the screw through at least 20 mm of available travel, then return to its starting position.
3. Check that neither end of screw or pad travel limits that stroke. Leave room to back the pads away after insertion.
4. Try both wooden handles in their intended positions. Both complete frames must clear the tube path, bench and neighbouring clamp.

CHECK THE REAL CLAMPS

The drawn pads and handles are nominal catalogue-form proxies. Check unloaded travel, full-face pad seating and handle access on the actual clamps before applying load; this setup supplies no force rating.

G supports the annular cap

The shallow recess locates the seal's cap while leaving its entire cylindrical wall accessible to your fingers.



The existing thimble over the puck's Ø11.2 mm, 1.8 mm-deep recess. The Ø6.8 mm through-hole leaves the seal's cap hole open.

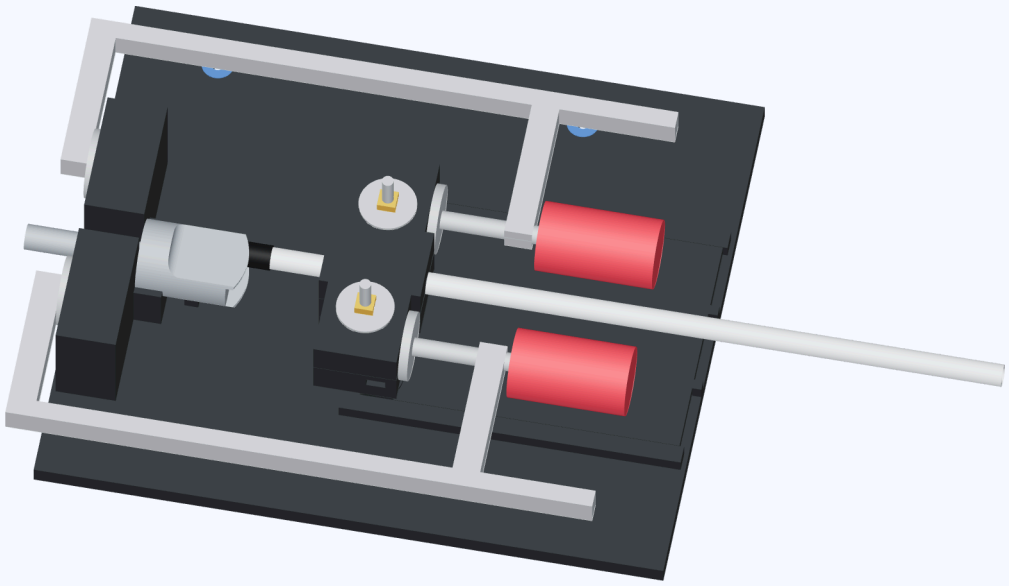
1. Clear the puck's through-hole and shallow recess of print remnants.
2. Set it recess-up on the bench. It receives the fresh thimble cap-down during sleeve fitting.
3. Keep the puck loose. It is a bench support, not a piece that enters the donor with the tube.

FINISHED WHEN

The recess is clean and the through-hole is open.

Leave both handles clear

Secure the base on a firm bench so both hands can turn the wooden screw handles. Keep the two metal frames in their seating plane.



Both frames open outward, with clear room for each wooden handle and the free tube. Blue identifies the base mounting locations.

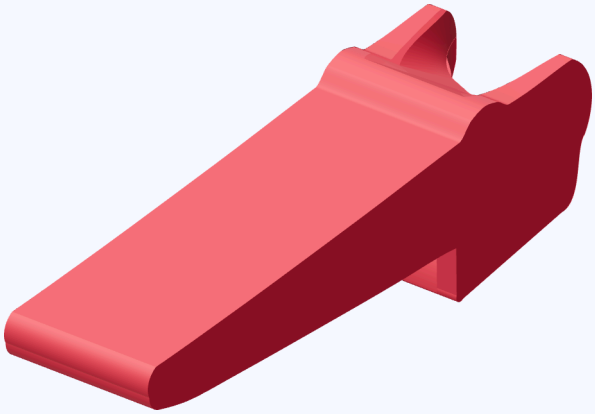
1. Set the base flat with room for the donor shank, both complete metal clamp frames and the free tube.
2. Use the two owned 4-inch steel C-clamps to hold the base at the bench edge, clear of all moving parts. Existing mounting bolts can also use the base holes.
3. Check handle access unloaded. Keep the tube and faucet lever out of both clamp frames and keep each screw pad aimed squarely at its rear grip contact.

FINISHED WHEN

The base stays stationary and both hands can reach and turn their handles without shifting a clamp or touching the tube.

Attach the faucet lever first

The bare Touch-Flo and its accepted faucet lever are already connected before the tube or the shell is fitted.



The accepted printed faucet lever, shown separately for identification. Its mounted rest position leaves the top water port accessible.

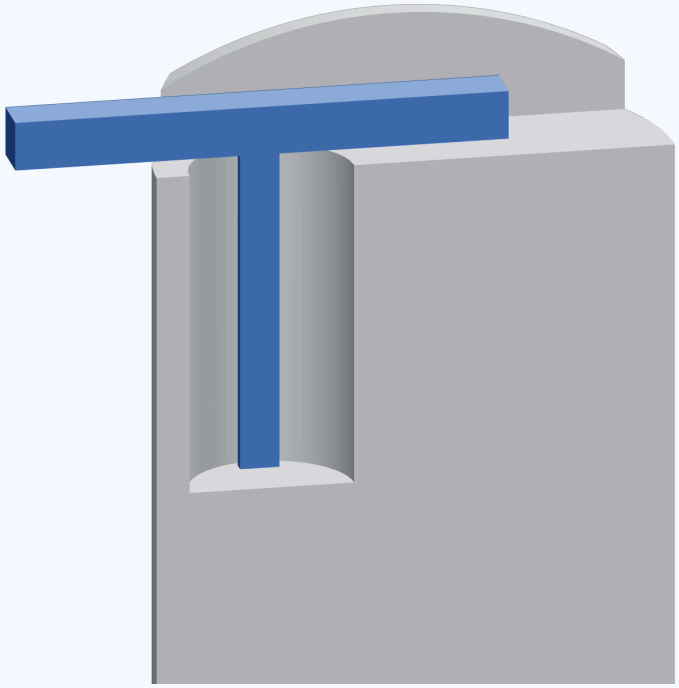
1. With the tube absent, lower the accepted faucet lever aft of its working position and slide it forward around the metal cylinder.
2. Leave the faucet lever **down at its unpressed rest**. Keep it attached for sleeve insertion and later shell assembly.
3. Confirm the donor's top port is empty. The thimble belongs on the tube before it enters this port.

FINISHED WHEN

The faucet lever is attached at unpressed rest and the top port is empty.

Measure the actual port depth

This reading locates the tube grip relative to the actual port floor. It sets the nominal travel limit without first fully seating the tube in the donor.



Caliper depth blade at the annular seating floor beside the central water opening. D is measured from the port mouth.

1. Use the existing caliper's depth blade. Rest its reference beam on the **port mouth below the raised arches**, clear of the plunger. An arch-top reading adds the wrong height.
2. Lower the blade gently beside the bore wall until it touches the **annular seating floor** that supports the thimble cap.
3. Read and record the depth as **D**, in millimetres. Check at the opposite side of the annular floor to confirm the same datum.
4. If the caliper cannot reach that datum, gently touch a clean 4 mm hex-key tip to the annular floor, mark the key at the port mouth, withdraw it and measure from the tip to the mark with the caliper.

MEASURE THE CAP SEAT

Keep the blade out of the central water opening and away from the valve plunger. The port's nominal 20 mm depth is not the fixture setup measurement.

Two marks from the tube end

Measure from the bare tube's square-cut end before the thimble hides it. The cap thickness is 2.1 mm.



Measure from the lower tube end shown. The lower mark is the port-lip witness; the upper mark locates the front of the jaws.

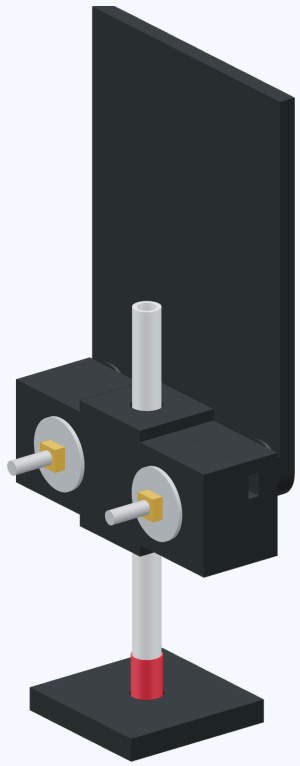
1. Confirm a square, clean tube end and an open bore. Use the tube length specified for the faucet finish.
2. Mark a fine line **D - 2.1 mm** from that end. This is the **port-lip seat witness**.
3. Mark a second line **D + 16.9 mm** from the same end. This is the **clamp-front mark**.
4. Keep the two lines distinguishable. The clamp-front line sits 19 mm farther up the tube than the port-lip witness.

FOR EXAMPLE ONLY

If the actual D reading is 20.0 mm, the lines lie 17.9 mm and 36.9 mm from the tube end. Use **your measured D**.

Work the thimble onto the tube

Lift the T handle and attached carrier out of the base. Use the grip and cap-support puck to fit the seal evenly outside the donor.



The puck supports the annular cap. The T handle holds the tube while the thimble wall advances evenly around it.

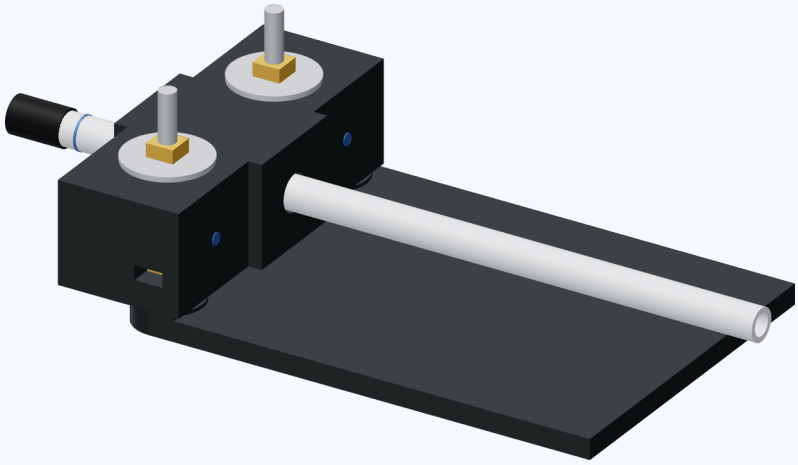
1. Grip the tube with the T handle far enough from the end to leave the entire thimble and both marks accessible.
2. Set the fresh TPU 85A thimble cap-down in the puck's shallow recess. Its open end faces the tube.
3. Center the square-cut tube end at the opening. Work the sleeve rim over it evenly around the circumference while advancing the tube.
4. Press both T-handle wings evenly in small increments. Pause tube advance while working the rim evenly around its circumference; keep the sleeve straight until the tube end contacts the annular cap.
5. Lift the sleeved tube off the cap-support puck. The puck stays on the bench for the donor insertion operation.

FINISHED WHEN

The sleeve wall is even, with no fold or roll, and the square-cut tube end rests on the cap. The cap hole remains open into the tube bore.

Move the grip to its mark

The grip pushes the tube. The thimble and port remain free of the jaws.



The jaw front aligns with the D + 16.9 mm tube mark. The two blue dots on the rear jaw seam mark the screw-pad centres, 18 mm inward from the outer wing edges.

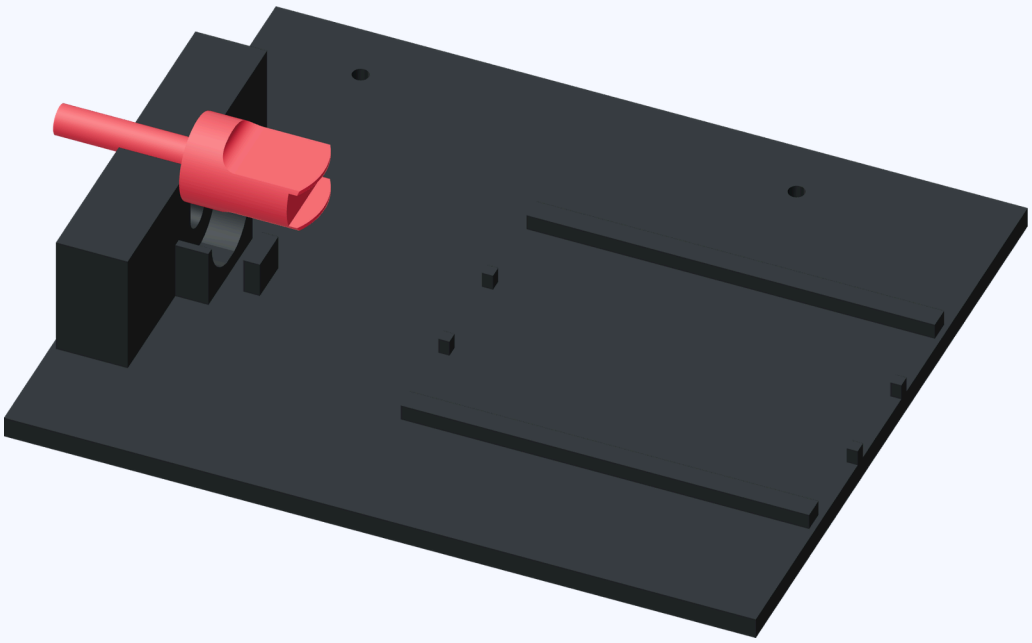
1. Loosen the two jaw screws enough to open the grip and relocate it without dragging a closed liner along the tube.
2. Place the two liners around the bare LLDPE with the **front edge of both jaws** at the clamp-front mark.
3. Close the rigid jaw shoulders evenly using the two M5 screws. The rigid closure shoulders limit compression; do not add a spacer or extra squeeze.
4. Confirm both marks are where intended and the tube end remains fully seated on the thimble's cap.

WATCH THE GRIP AND TUBE

Use the designed jaw closure. If the grip slips during insertion, or the tube stays flattened after release, stop and inspect the liner fit and jaw closure.

Place the donor in the nest

The horizontal fixture supports the robust cylindrical metal base. The attached faucet lever remains in the open front-side clearance.



The shank exits left and the water port faces the tube. The attached faucet lever is omitted from this CAD view; keep it down at unpressed rest on the open front side.

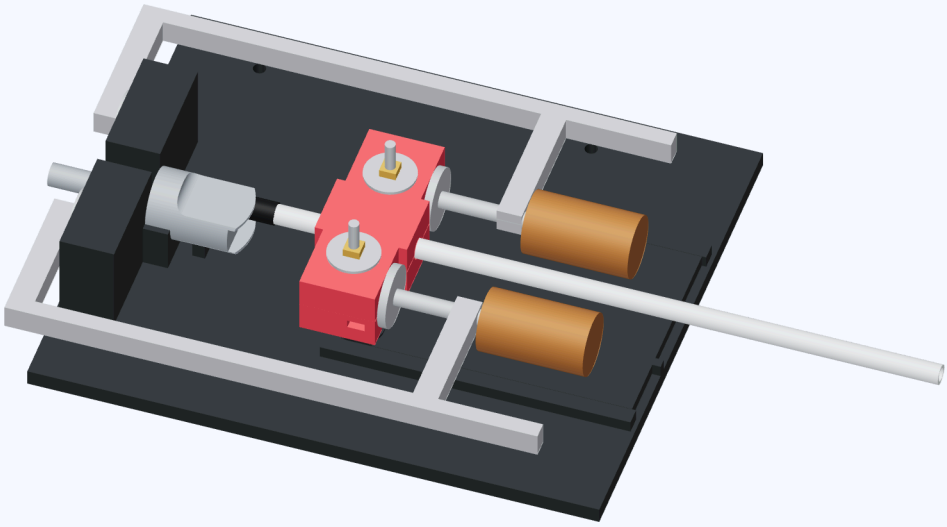
1. Lay the bare Touch-Flo in the nest with its cylindrical base in the saddle and its flat bottom annulus against the donor reaction block.
2. Let its threaded shank pass through the open notch. Rest the flat metal column on the clocking pad so the empty top port aligns with the tube axis.
3. Keep the attached faucet lever **down at unpressed rest** in the open side facing you. The fixture must touch neither the lever nor the valve plunger.

FINISHED WHEN

The metal body rests against the donor reaction block, the shank is clear and the unpressed faucet lever has open clearance.

Seat both pressure points

The common carrier guides the grip. Both fixed pads and both screw pads must seat full-face before either screw advances under load.



Both metal frames are complete. Their screw pads meet the two rear grip faces at tube height; fixed pads meet the rear of the donor reaction block. The faucet lever is omitted; keep it attached down at unpressed rest.

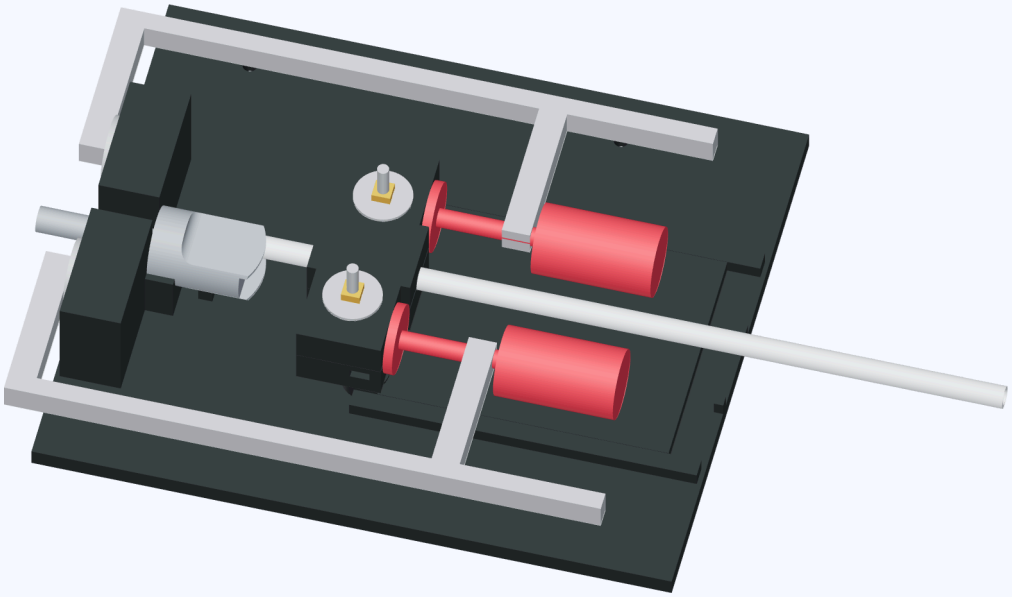
1. Set the sleeved cap at the port and align the tube with its bore. Keep B attached to D; lift the assembly clear of the guides for any initial hand advance.
2. For **D > 20 mm**, press both T wings evenly to start the sleeve **D - 20 mm** into the port. Stop when B can enter the fully retracted guide position.
3. For **D ≤ 20 mm**, keep the cap at the mouth and place B between its guides **20 - D mm** ahead of the rear start. The loaded travel is D.
4. Confirm the B-to-D joint remains seated. Fit C1/C2 at the marked pad regions, with their frames open outward in the same plane.
5. Bring both screws into light contact. Check all four pad faces, grip position, donor seating and free faucet-lever clearance.

FINISHED WHEN

All four factory pads seat fully, both screws contact the common grip, and the tube stays straight with the grip front at its mark.

Advance in small steps

Use both hands on C1/C2's compact wooden handles. Advance the common grip in small steps and watch that it stays square.



Two screw-pad stations act on one grip. A small advance can open a gap at the opposite pad. The faucet lever is omitted from this nominal view; keep it attached down at unpressed rest.

1. Bring both screws into light contact with their rear grip faces. Confirm the fixed pads remain seated and the donor annulus stays against the donor reaction block.
2. Advance both handles together when convenient. Otherwise make one very small advance, then immediately take up the opposite pad's gap before another advance. Watch that the grip stays square and at its tube mark.
3. Stop at the **first** port-lip witness or travel-stop contact. If they disagree, unload both screws and recheck the depth, marks, sleeve, grip and alignment.
4. Unload and inspect if a gap or skew grows, a pad tilts, a part flexes, the tube bows or slips, the sleeve folds, or an advance produces no tube movement.

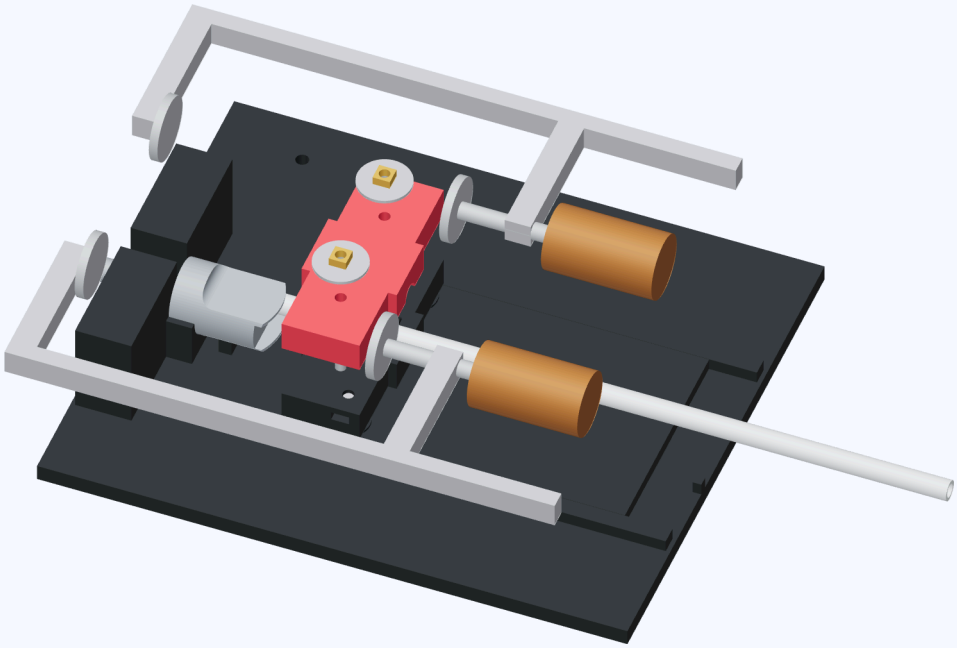
RELEASE WHEN MOTION CHANGES

Keep the clamps in one plane. Do not rock the tube or continue increasing effort against binding or a stop. The stop limits travel; it does not prove full seating or load capacity.

AFTER INSERTION · OPERATION

Unload both clamps first

Release the two screw loads before opening the grip. Keep the tube and donor supported together.



Both complete clamps are unloaded and lifted clear before E opens. The bolted B/D assembly remains in the base. The faucet lever is omitted; keep it attached at unpressed rest.

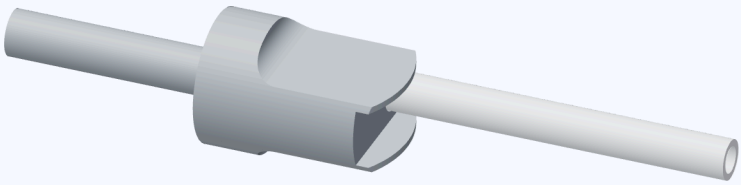
1. Back off each screw a little in turn until both moving pads are unloaded and clear of the rear grip faces.
2. Support the grip while taking the two metal clamps off their pad regions. Keep the tube and donor together in the nest.
3. Loosen the two M5 × 50 grip screws and remove E. Leave the bolted B/D assembly in the base with its lower channel open.
4. Lift the donor and seated tube together out of the nest and lower liner. Check the seating witness and released grip area: the tube must remain round, without a persistent flat, gouge or folded sleeve.

FINISHED WHEN

Both metal clamps and the grip are removed, the tube retains its witness and the faucet lever remains attached down at unpressed rest.

Carry the assembly to the shell

The tube, thimble and faucet lever remain attached to the Touch-Flo. The insertion tooling comes away.



The donor and sleeved tube after removal from the jig. The faucet lever is omitted from this CAD view; keep it attached for shell entry.

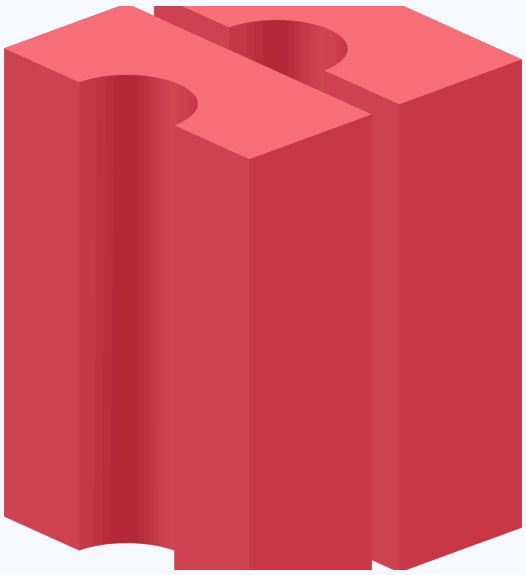
1. Follow the matching faucet assembly guide for the tube routing and shell entry.
2. Flip the attached faucet lever up to its entry position before inserting the donor into the shell.
3. As the donor seats, the shell pushes the faucet lever forward. Once fully seated, the shell prevents that flipped-up position.
4. Allow the faucet lever to snap down to its **unpressed rest** position and check its normal actuation and return.

FINISHED WHEN

The donor is seated, the tube retains its sleeve and seating witness, and the faucet lever rests down without being pressed.

Keep the contacts clean

The printed fixture is an unprinted prototype. The observed faucet sequence is established; grip, insertion force and fixture endurance remain unmeasured.



Replaceable F liners have smooth half-round channels and full flat backings. Use the prepared liner project when a liner is damaged.

1. Remove strings and debris from liners, guide faces, pad regions, stops and the donor nest. Keep the cap-support puck hole clear.
2. Replace a torn, rolled or permanently distorted liner. Inspect the printed jaw contact faces, guides and donor reaction block for cracking or permanent deformation.
3. Keep both metal clamp pads clean and able to seat flat. Inspect the bars, sliding jaws, screws and swivel pads before use.
4. Store the grip and liners unloaded, with the metal clamp screws backed off.

CAD and native slicing establish nominal shape and print preparation. They do not establish grip retention, full seating, equal load sharing, fixture strength or lifetime. The supplier's clamp specifications describe the clamp, not a rating for this assembly.

Fixture source and ready print files
Observed faucet assembly sequence